

Model-Relative Bias of Omega Under Equivalent Measurement Models

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Purpose

Bell, Chalmers, and Flora [1] define population reliability as the omega parameter attached to the model that generated the covariance matrix. They then call an estimated omega biased when its expectation differs from that model-relative target. That is internally coherent.

The problem is that the target is not, in general, a functional of the observed item distribution. Equivalent latent decompositions can imply the same covariance matrix while assigning different amounts of composite variance to the named target factor. In that case the same omega estimate has different population bias under different, observationally indistinguishable, “true” models. This note gives a simple theorem and a numerical example where the bias sign changes without changing the observable covariance matrix.

1 Model-Relative Omega

Let $y \in \mathbb{R}^p$ be the item vector and let $X = w^\top y$ be the composite, usually with $w = \mathbf{1}$. Write $\Sigma = \text{Var}(y)$. If a target factor f with $\text{Var}(f) = 1$ contributes the item loading vector a in a decomposition

$$y = af + u, \quad \text{Cov}(f, u) = 0, \quad \text{Var}(u) = \Psi,$$

then the model-relative omega target for the composite X , with respect to the named factor f , is

$$\omega(a, \Sigma; w) = \frac{\text{Var}(w^\top af)}{\text{Var}(w^\top y)} = \frac{(w^\top a)^2}{w^\top \Sigma w} = \frac{(w^\top a)^2}{(w^\top a)^2 + w^\top \Psi w}. \quad (1)$$

The denominator is the full composite variance, including residual covariances. It is not obtained by replacing Ψ with $\text{diag}(\Psi)$, unless the residual covariance matrix is actually diagonal.

For a one-factor model, a is the loading vector. For a bifactor model, a is the general-factor loading vector. For a higher-order model, a is the vector of indirect effects from the higher-order factor to the items. The Bell et al. bias calculation is therefore

$$\mathbb{E}(\hat{\omega}) - \omega(a_{\text{true}}, \Sigma; w),$$

where a_{true} is supplied by the chosen population model.

The word “chosen” matters. The covariance matrix alone need not determine a_{true} .

2 A Non-Identification Result

Theorem 1 (Equivalent models with arbitrary composite omega). *Let Σ be positive definite and let $w \neq 0$. For every $r \in [0, 1]$, define*

$$a_r = \sqrt{r} \frac{\Sigma w}{\sqrt{w^\top \Sigma w}}, \quad \Psi_r = \Sigma - a_r a_r^\top. \quad (2)$$

Then $\Psi_r \succeq 0$, the one-factor representation

$$y = a_r f_r + u_r, \quad \text{Var}(f_r) = 1, \quad \text{Cov}(f_r, u_r) = 0, \quad \text{Var}(u_r) = \Psi_r$$

implies the same covariance matrix Σ . In that representation, the omega of the composite $X = w^\top y$ with respect to the common factor f_r is

$$\omega(a_r, \Sigma; w) = r.$$

If all entries of Σw are nonzero, the target factor loads on every item for every $r > 0$.

Proof. The covariance equality is immediate:

$$\text{Var}(a_r f_r + u_r) = a_r a_r^\top + \Psi_r = \Sigma.$$

For any vector x , Cauchy’s inequality in the inner product $\langle x, z \rangle_\Sigma = x^\top \Sigma z$ gives

$$(x^\top \Sigma w)^2 \leq (x^\top \Sigma x)(w^\top \Sigma w).$$

Therefore

$$x^\top \Psi_r x = x^\top \Sigma x - r \frac{(x^\top \Sigma w)^2}{w^\top \Sigma w} \geq (1 - r) x^\top \Sigma x \geq 0,$$

so $\Psi_r \succeq 0$. Finally,

$$(w^\top a_r)^2 = r w^\top \Sigma w,$$

which proves $\omega(a_r, \Sigma; w) = r$. $\square$

Corollary 1 (Bias sign is not identified). *Let an omega estimator converge under the observed covariance Σ to some value $T(\Sigma) \in (0, 1)$. In the equivalent model family from Theorem 1, its asymptotic bias is*

$$T(\Sigma) - r.$$

Thus, without changing the observed covariance matrix, one can choose an equivalent population model under which the estimator is positively biased, unbiased, or negatively biased.

The corollary is the clean version of the objection. Bias is identifiable only after fixing a target factor by theory or by a restrictive model class that makes the target unique. It is not identified by the covariance matrix itself.

3 Numerical Example

Take $p = 8$, $w = \mathbf{1}$, and the compound-symmetry correlation matrix

$$\Sigma_\rho = (1 - \rho)I + \rho \mathbf{1}\mathbf{1}^\top, \quad \rho = .30.$$

The independent-error one-factor decomposition has

$$a_0 = \sqrt{\rho} \mathbf{1}, \quad \Psi_0 = (1 - \rho)I.$$

Its omega is

$$\omega_0 = \frac{p^2 \rho}{p + p(p - 1)\rho} = 0.774194.$$

A correctly fitted one-factor omega also converges to $T(\Sigma_\rho) = 0.774194$.

But the same Σ_ρ has the following equivalent one-factor representations with correlated residuals:

r	a_r entry	Ψ_r diagonal	Ψ_r off diagonal	$T(\Sigma_\rho) - r$
.600000	.482183	.767500	.067500	.174194
.774194	.547723	.700000	.000000	.000000
.850000	.573912	.670625	-.029375	-.075806

All three rows generate exactly the same item covariance matrix. The first says the same fitted omega is positively biased by about .17. The second says it is unbiased. The third says it is negatively biased by about .08. The only thing that changed is the latent story used to split Σ_ρ into target-factor variance and residual covariance.

4 Connection to Bell et al.'s Population Families

Bell et al. use four population families. With $d = w^\top \Sigma w$, their target vector a and target omega can be written as follows:

$$\begin{aligned} \text{simple one factor: } \Sigma &= \lambda \lambda^\top + \Theta_D, & \omega_u &= (w^\top \lambda)^2 / d, \\ \text{one factor with correlated errors: } \Sigma &= \lambda \lambda^\top + \Psi_G, & \omega_u &= (w^\top \lambda)^2 / d, \\ \text{bifactor: } \Sigma &= g g^\top + S S^\top + \Theta_D, & \omega_H &= (w^\top g)^2 / d, \\ \text{higher order: } \Sigma &= (\Lambda \gamma)(\Lambda \gamma)^\top + \Lambda \Psi_z \Lambda^\top + \Theta_D, & \omega_{ho} &= \{w^\top (\Lambda \gamma)\}^2 / d. \end{aligned}$$

Here Θ_D is diagonal residual variance, Ψ_G is a residual covariance matrix with Bell et al.'s selected local-dependence graph, S holds the bifactor specific-factor loadings, and $\Lambda \gamma$ is the vector of indirect effects from the higher-order factor to the items.

This display makes the target-dependence visible.

Corollary 2 (Bell's multidimensional covariances have one-factor correlated-error alternatives). *Let Σ be any covariance matrix generated by Bell et al.'s correlated error, bifactor, or higher-order population models. For every $r \in [0, 1]$, there is a one-factor model with correlated residuals, a common factor loading on every item whenever Σw has no zero entry, and the same observed Σ , for which the omega of $X = w^\top y$ with respect to that model's common factor equals r :*

$$\frac{\text{Var}(w^\top a_r f_r)}{\text{Var}(w^\top y)} = \frac{(w^\top a_r)^2}{w^\top (a_r a_r^\top + \Psi_r) w} = r.$$

Proof. Apply Theorem 1 to the covariance matrix generated by the Bell population model. $\square$

Corollary 2 is not a claim that Bell et al. secretly fit this alternative model. It is a claim about what the observed covariance identifies. A bifactor population covariance

$$\Sigma = g g^\top + S S^\top + \Theta_D$$

is also a one-factor correlated-residual covariance

$$\Sigma = a_r a_r^\top + \Psi_r$$

for every target reliability r . The same is true of a higher-order population after writing its target contribution as $a = \Lambda\gamma$. Thus the Bell population labels .60 and .85 are not labels for the covariance matrix alone. They are labels for a particular latent split of that covariance.

The four Bell families are therefore affected differently.

1. In the simple one-factor model with independent errors, the target is comparatively well anchored. With at least three indicators and nonzero, sign-consistent off-diagonal covariances, $\lambda\lambda^\top$ is identified from the off-diagonal part of Σ , up to a global sign. Since $(w^\top\lambda)^2$ is invariant to that sign, ω_u is identified inside this restricted model class. The equivalence objection does not attack Bell et al.’s correctly specified simple one-factor cells.
2. In the one-factor correlated-error model, the target is conditional on the selected residual covariance graph and on the decision that those residual covariances, rather than the common factor, carry the extra covariance. If that graph is treated as fixed and the model is identified, Bell et al.’s target is again well-defined conditionally. If the graph is part of the uncertainty, or if the analyst is only committed to “one common factor plus local dependence,” Theorem 1 applies directly. The same covariance can be assigned a lower or higher target omega by moving covariance between $\lambda\lambda^\top$ and Ψ_G .
3. In the bifactor model, Bell et al.’s target is the contribution of the direct general factor g . Marginally, however, the specific factors $SS^\top$ are just structured residual covariance relative to the general factor. A covariance generated by $gg^\top + SS^\top + \Theta_D$ can be re-expressed as a one-factor correlated-error model with any target omega r . Thus a statement such as “ $\hat{\omega}_u$ is biased relative to the true bifactor omega” is best read as “biased relative to the declared bifactor target,” not as a covariance-only fact.
4. In the higher-order model, the situation is less damaging for Bell et al.’s comparison between ω_H and ω_{ho} . A Schmid-Leiman restricted bifactor representation can preserve the higher-order target vector $a = \Lambda\gamma$.

When it does, the numerator in (1) is the same and ω is target-preserving across those two representations. The equivalence objection returns only when the fitted or alternative bifactor representation changes the general-factor contribution vector.

This also sharpens the role of model fit in Bell et al.’s RQ4. Exact equivalent models have the same fit to Σ . Fit indices can help reject visibly bad approximations, but they cannot choose among target-changing equivalent decompositions.

5 What This Does and Does Not Refute

The theorem does not show that Bell et al. made a computational mistake. Once their population model and target factor are stipulated, their Monte Carlo bias quantity is a legitimate model-relative quantity.

It does show that the broader language “ ω is biased for reliability under misspecification” needs a qualification. If reliability means the proportion of composite variance attributable to a substantively named factor, then that factor is part of the estimand. Equivalent measurement models that change the factor contribution vector a change the estimand while leaving the data distribution fixed. No sample size or fit index can recover which bias statement is the correct one from Σ alone.

There is also an important non-objection. Some equivalent representations preserve the target contribution vector. For example, the Schmid-Leiman transformation relates restricted bifactor and higher-order models [2, 3]. When the bifactor general-factor loadings equal the higher-order indirect effects, the numerator in (1) is the same, so ω is invariant. Equivalence is damaging only when it is not target-preserving.

6 Implication for Simulation Design

A simulation can still answer a conditional question:

If this latent model is the true target story, what does this fitted ω estimate?

It cannot, by itself, answer an unconditional question:

How biased is ω for the reliability encoded in the observed covariance matrix?

There is no such covariance-only reliability target unless additional identifying restrictions are imposed.

For future reliability simulations, the defensible formulation is therefore:

Bias is relative to a declared target model. Before interpreting it as a failure of an estimator, show that the target is invariant across the relevant equivalent models, or explicitly treat the result as a sensitivity analysis over plausible target decompositions.

Under that standard, the user’s objection is right in the strong sense: there are equivalent “true” models under which the same fitted omega has smaller, larger, zero, or opposite-signed bias. The objection does not make Bell et al.’s conditional Monte Carlo invalid, but it blocks a covariance-only or model-free reading of their bias claims.

References

- [1] Bell, S. M., Chalmers, R. P., & Flora, D. B. (2024). The impact of measurement model misspecification on coefficient omega estimates of composite reliability. *Educational and Psychological Measurement*, 84(1), 5–39. doi:10.1177/00131644231155804.
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